

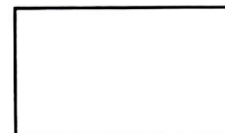


# East West University

Department of MDS

MAT 205

Section: 8



Marks Obtained

Name: .....

Quiz No.: 03

Date: 17.08.2026

ID. No.: .....

Topics: Vector Space and Complex Analysis

Total Marks: 20

Time: 25 minutes

1. Write the first fundamental theorem of subspace. Express the vector  $\underline{v}$  as a linear combination of the vectors  $\underline{v}_1, \underline{v}_2$ , where

$$\underline{v} = (2, -5, 4) \text{ and } \underline{v}_1 = (1, -3, 2), \underline{v}_2 = (2, -1, 1). \quad 5$$

2. Verify whether the function  $f(z) = \frac{1}{z}; z \neq 0$  is analytic. Is it entire? Justify your answer. . . . 6

3. For the function  $f(z)$ , locate and name all the singularities in the finite  $z$ -plane, where

$$f(z) = \frac{z^8 + z + 2}{(z-1)^3(3z+2)^2}. \text{ Discuss briefly the essential and removable singularities in relation to the given function.} \quad 9$$